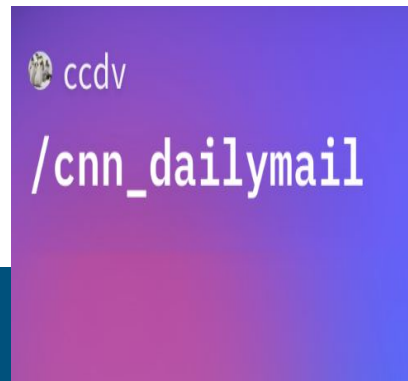
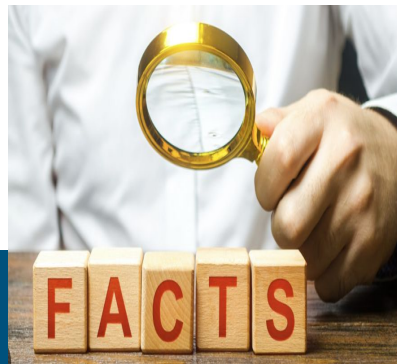
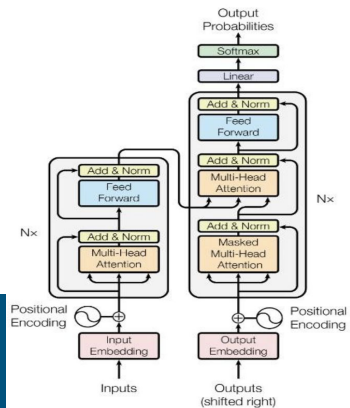


When Metrics Say “Yes” But Humans Say “No”

*A Case Study of Hallucination Reduction for Factual
Summarization*

Author: Ryan Powers

Project Characters



BART(Generator)

Encoder-decoder,
140M params

FactCC(Verifier)

Binary classifier
(Consistent/Inconsistent)

CNN/Daily Mail
(Dataset)

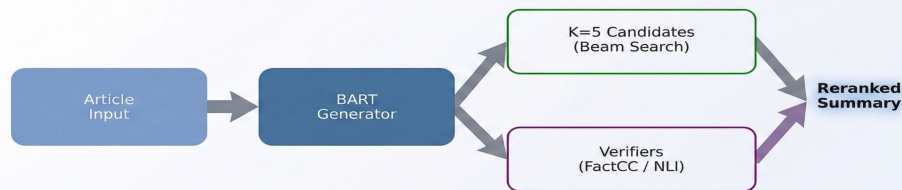
287K train / 13K val /
11K test

Single Human
Annotator

Blind comparison,
stratified sample

Task/Problem/Goal

Project Architecture: Verifier-Reranking Pipeline



Can I just download a standard fact-checker and use it to fix my model's errors?

Task:

Test whether better automatic metrics $\Rightarrow$ better human-judged summaries.

Problem:

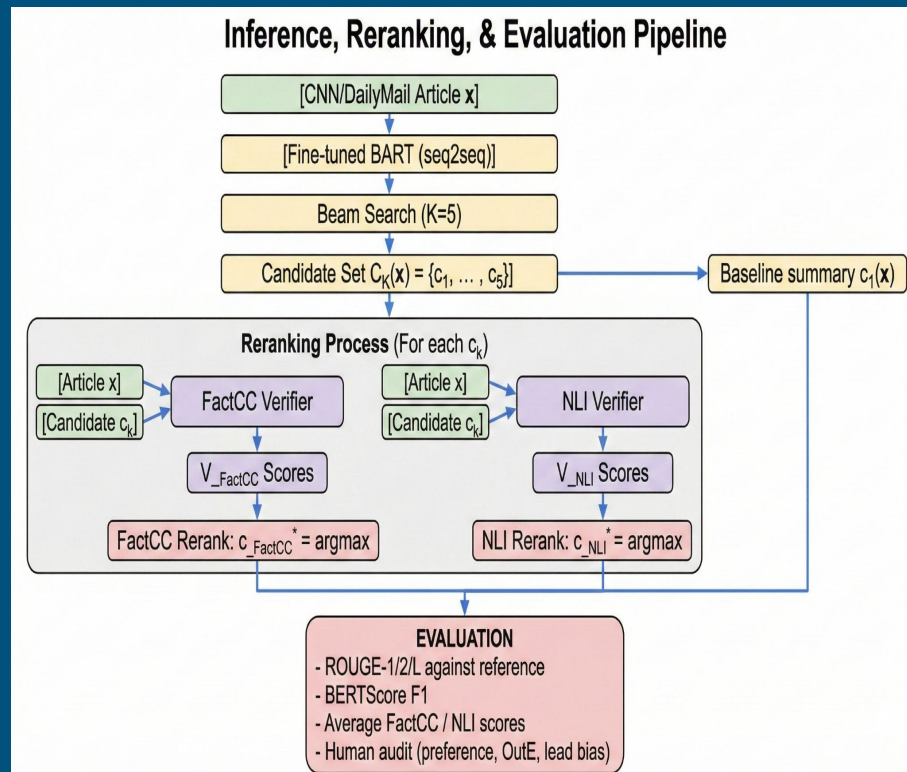
Summaries are fluent but sometimes factually wrong (hallucinations)

Goal:

Improve factual consistency **without** retraining a bigger model, use reranking instead.

Architecture Diagram (Generate → Verify)

- Generate **K=5** candidates with BART (beam search)
- Score each candidate with:
 - **FactCC**: probability the summary is consistent with the article.
 - **RoBERTa-MNLI**: entailment score (for ablation).
- Pick the candidate with highest FactCC score.



BART + FactCC / Experimental Setup

Models

- **Generator:** facebook/bart-base (fine-tuned on 20k CNN/DM, 3 epochs, lr=5e-5) -
 - a. **Verifier A:** FactCC (factual consistency) -
 - b. **Verifier B:** RoBERTa-MNLI (ablation)

Data

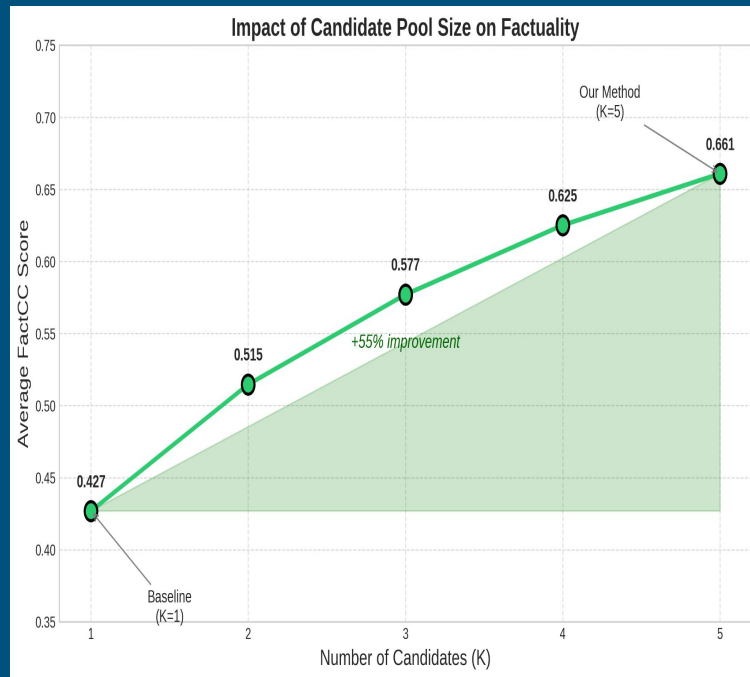
- **Train:** 20k subset (seed=42) - **Val:** 2k | **Test:** 11,490

Decoding

- **Beam search, K=5, max length 140**

Success Criteria ✓ $\Delta\text{FactCC} \geq +2.0$ points ✓ ROUGE-L drop ≤ 1.0 point

After analyzing K-ablation shows strong gains up to ~3–5, with diminishing returns.



Automatic Results

- **ROUGE-L:**
 - **LEAD-3: 24.9**
 - **BART baseline: 28.09**
 - **FactCC rerank: 27.92**
 - **NLI rerank: 28.08**
- **FactCC (avg factuality score):**
 - **Baseline: 0.427**
 - **FactCC rerank: 0.661**
(+55%, $p < 0.00001$)
- **Success criteria satisfied:**
 - **$\Delta\text{FactCC} \geq +2.0$ ✓**
 - **ROUGE-L drop ≤ 1.0**
(-0.17) ✓

Table 1: Main results on CNN/DailyMail test set ($N=11,490$). R-1/2/L = ROUGE F_1 ($\times 100$). BS = BERTScore F_1 . FactCC = mean factuality score. Best factuality in **bold**.

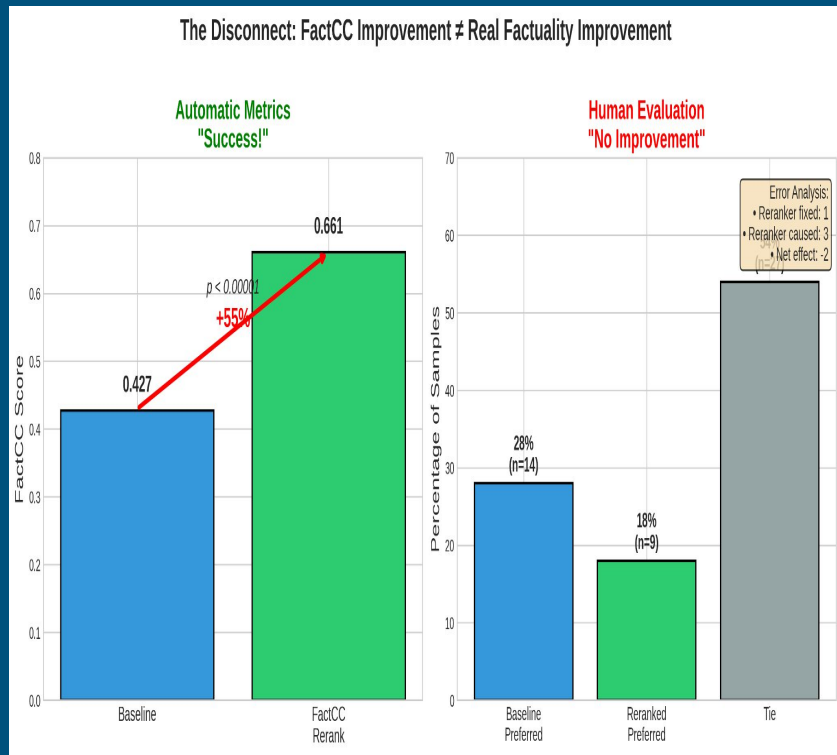
System	R-1	R-2	R-L	BS	FactCC
LEAD-3	39.91	17.38	24.91	—	—
BART Baseline	41.12	18.56	28.09	0.882	0.427
FactCC Rerank	40.74	18.36	27.92	0.882	0.661
NLI Rerank	40.78	18.48	28.08	—	—

Δ (FactCC Rerank vs Baseline): FactCC +0.234 (+55%, $p < 0.0001$), R-L -0.17 (n.s.)

Table 2: Impact of candidate pool size (K) on factuality. Marginal gain shows diminishing returns beyond $K=3$.

K	Avg FactCC	Δ from $K=1$	Marginal Gain
1	0.427	—	—
2	0.515	+0.088	+0.088
3	0.577	+0.150	+0.062
4	0.625	+0.198	+0.048
5	0.661	+0.234	+0.036

Human Audit: 50 - Example Evaluation



- Stratified Sampling:
 - 30 where baseline vs rerank disagree
 - 20 where they agree
- Human (Single Annotator) preference:
 - Tie: 27 (54%)
 - Baseline preferred: 14 (28%)
 - Reranked preferred: 9 (18%)
- Net hallucination effect: reranker fixes 1 error, introduces 3 → **Net -2**

Error Taxonomy (Human-Labeled, $N = 50$)

Error Type	Definition	Count (N=50)
None	Summary contains no factual errors	34 (68%)
Entity	Wrong name, number, place, or factual attribute	0 (0%)
Temporal	Incorrect time, date, or sequence of events	0 (0%)
Contradiction	Summary directly contradicts the article	0 (0%)
Out-of-Context	Fabricated or unsupported factual claim ("hallucination")	16 (32%)

When Reranking Helps vs Hurts

The Success (Logic Repair)

- **Context:** Refugees formed a human chain to stop attackers.
- **Baseline (The Mistake):** "15 men arrested... But were *prevented* [from being arrested] because they huddled together..."
 - Status: **Logical Error** (Implies the chain stopped the police).
- **Reranked (The Fix):** "...claimed the men tried to throw other Christians off... but were *prevented* [from doing so] because they huddled together..."
 - Status: **Corrected** (The verifier caught the causal mismatch).

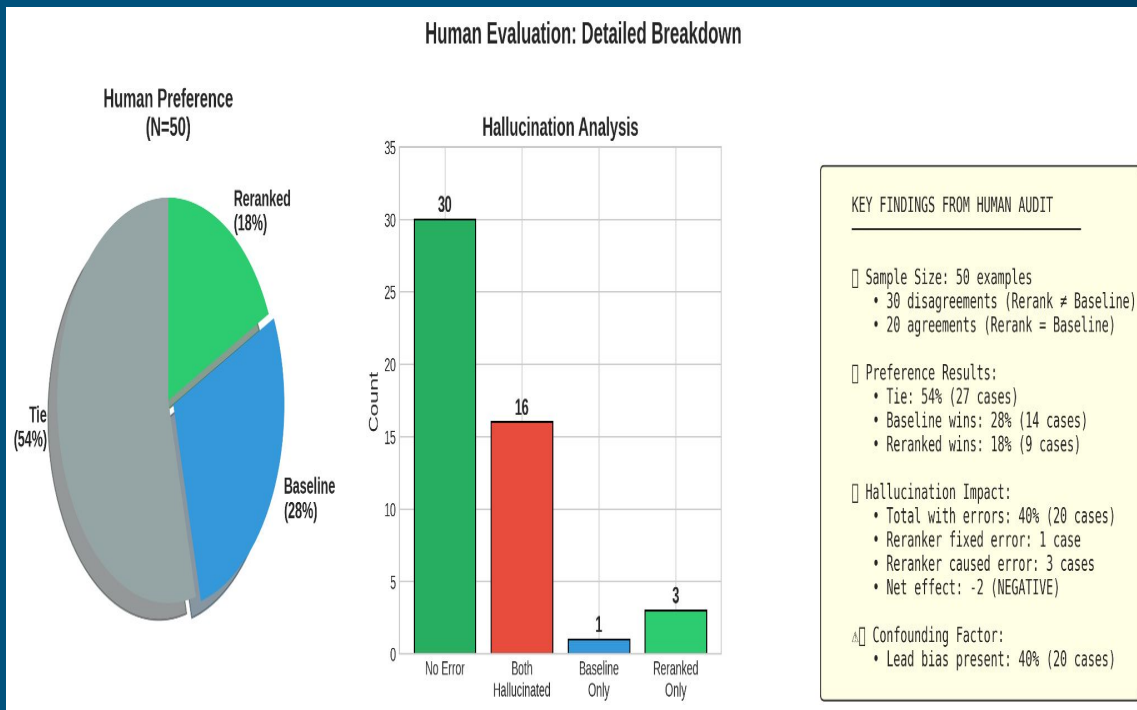
The Failure (Hallucination)

- **Context:** A father hit his toddler; store owner gave video to police.
- **Baseline (The Safe Bet):** "Justin Whittington... was taken into custody... relationship not yet revealed..."
 - Status: **Boring but Factual** (Stuck to the text).
- **Reranked (The Lie):** "...Police were notified... and asked the public for help identifying the man."
 - Status: **Hallucination** (The article never mentioned a public appeal).

Verifier (FactCC) isn't actually checking facts it's just checking for patterns.

Conclusion

FactCC improved +55% ($p < 0.00001$), but human audit showed 0% correction rate and net -2 hallucinations.



Low candidate diversity: Beam search produces similar candidates (84% pairwise similarity)

Verifier-human misalignment: FactCC prefers extractive, lead-like summaries that humans find less informative

Lead bias in CNN/DM: LEAD-3 achieves 24.9 ROUGE-L only 3.2 points below BART so "safe" extractive choices score well

- **Why:** Goodhart's Law. Once I made FactCC the optimization target, the system learned to game FactCC rather than be more truthful.

References & Github

- Lewis et al. 2020. “BART: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation.” *ACL*.
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- Ji et al. 2023. “Survey of Hallucination in Natural Language Generation.”
- Zha et al. 2023. “How Well Do Factuality Metrics Align with Human Judgments?” (for your QAGS/FactCC alignment discussion).

<https://github.com/rPowDS/MIDS266-NLP-final-project>